

Cross-Section Reconstruction via Truncated SVD: A Drop-In Replacement for Pointwise Tables in Monte Carlo Neutron Transport

F. Rodrigues

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Abstract

Continuous-energy Monte Carlo neutron transport codes store cross sections as pointwise tables indexed by energy and temperature, reading them through binary search plus log-log interpolation. We replace this lookup with a truncated singular value decomposition (SVD) of $\log_{10} \sigma(E, T)$: at evaluation time each reaction channel reduces to a rank- k fused multiply-add over contiguous memory. We implement both providers—pointwise table and SVD reconstruction—inside a single pure-Rust Monte Carlo engine with full physics (anisotropic elastic scattering, tabulated fission spectra, URR probability tables, $S(\alpha, \beta)$ thermal scattering, discrete inelastic levels, energy-dependent $\bar{\nu}(E)$), so that changing provider changes only the cross-section lookup. The same two providers are available in an event-based CUDA transport solver built on the Tramm architecture [10]. Five-seed criticality benchmarks on Godiva (HEU-MET-FAST-001) and a standard PWR pin cell with 9 nuclides yield four observations. (i) The singular spectrum of ENDF/B-VII.1 cross sections decays rapidly ($\sigma_2/\sigma_1 = 7.7 \times 10^{-2}$ for ^{235}U fission); 47 of 52 ^{235}U reaction channels are effectively rank-1. (ii) Rank-6 reconstruction reaches machine precision at 44 ns per lookup against 91 ns for a binary-search pointwise table, a $2.0\times$ kernel-level speedup. (iii) On the PWR pin cell, SVD rank-5 is statistically indistinguishable from the pointwise table in the same engine (both within 291 pcm of the OpenMC reference) and runs end-to-end at 28 651 ns/particle against 54 430 ns/particle for the pointwise table: a $1.90\times$ full-transport speedup at matched fidelity. (iv) The GPU solver reproduces OpenMC k_∞ within one standard error on both pointwise and SVD paths, confirming physics parity. A hybrid SVD + Windowed Multipole architecture is outlined as the natural complement for the resolved resonance region, with $530\times$ memory reduction per reaction channel (15 KB against 7.8 MB). SVD is not a replacement for every data structure in nuclear transport, but at production-relevant ranks it is a drop-in substitute for the pointwise table that runs faster without measurable loss of fidelity.

Keywords: Monte Carlo, cross-section compression, singular value decomposition, nuclear data, GPU transport.

1 Introduction

Continuous-energy Monte Carlo neutron transport codes such as OpenMC [1], MCNP, and Serpent rely on pointwise cross-section tables. Each reaction channel is sampled on a piecewise-linear union grid spanning 10^4 – 10^5 energy points per nuclide and temperature, and a production full-core calculation may reference several hundred nuclides at several temperatures for Doppler feedback. The aggregate data volume reaches tens of gigabytes, far exceeding the capacity of modern CPU caches. Every cross-section lookup during particle transport—binary search on a large monotonic array followed by log-log interpolation between two adjacent values—is memory-bound, and the access pattern (one random cache line per thread per lookup) also defeats GPU warp coalescing. Prior work has attacked this bottleneck along several axes: machine-learned resampling retains 10–50% of the data with ≤ 97 pcm criticality error on reactor benchmarks [2,

3]; the Windowed Multipole method gives an analytical form for the resolved resonance region [4, 5, 6]; and recent GPU-centric architectures reorganise the transport loop to amortise the lookup across warps [10, 11].

This paper takes a different route. We do not compress the data by dropping points; we replace the representation. The $\log_{10}$ -transformed cross-section matrix $\mathbf{A} \in \mathbb{R}^{N_E \times N_T}$ indexed by energy and temperature is approximated by its rank- k singular value decomposition, $\log_{10} \mathbf{A} \approx \mathbf{U}_k \mathbf{\Sigma}_k \mathbf{V}_k^\top$. At evaluation time a single cross-section value reduces to a k -wide dot product between one row of the pre-multiplied basis and a column of temperature coefficients. For representative ENDF/B-VII.1 nuclides the singular values decay rapidly— $\sigma_2/\sigma_1 \approx 8\%$ for ^{235}U fission, and 47 of 52 ^{235}U reaction channels are within machine epsilon of rank one—so $k = 5\text{--}6$ is ample for production fidelity.

We implement both providers, the traditional pointwise table and the SVD reconstruction, inside a single pure-Rust Monte Carlo transport engine with full physics, and inside a matching event-based CUDA solver. A command-line flag selects the provider. All other code paths—geometry, collision physics, RNG, angular distributions, fission spectra, URR tables, thermal scattering, discrete inelastic levels—are bit-identical between the two modes. Any k_∞ or timing difference between the two providers is therefore an artefact of the cross-section lookup alone. This is the setup we use to quantify the fidelity and throughput of SVD reconstruction as a drop-in replacement for the pointwise table.

Contributions.

- A cache-resident rank- k SVD reconstruction kernel for pointwise cross-section data, with temperature interpolation by the Ducru kernel method [8] (Sec. 2).
- A measurement of the per-lookup cost of the kernel against a binary-search pointwise table on the same energy grid: rank-6 reaches machine precision at 44 ns against 91 ns for the table (Sec. 4.2).
- A Godiva (HEU-MET-FAST-001) rank sweep showing that all ranks $k \geq 2$ agree with the pointwise table in the same engine to within one σ_{seed} (Sec. 4.3).
- A PWR pin cell rank sweep (9 nuclides, $S(\alpha, \beta)$, URR, discrete levels) showing CPU SVD rank-5 at 28 651 ns/particle against CPU table at 54 430 ns/particle, $1.90\times$ faster at matched fidelity and within 291 pcm of OpenMC 0.15.3 (Sec. 4.4).
- An event-based CUDA transport solver built on the Tramm architecture [10] implementing both providers; both match OpenMC k_∞ within a single standard error at rank-5 on the PWR pin cell (Sec. 4.5).
- A hybrid SVD + Windowed Multipole architecture for the resolved resonance region with $530\times$ memory reduction per reaction channel relative to the pointwise table (Sec. 4.6).

2 Method

2.1 Problem formulation

Let $\sigma(E, T)$ denote the microscopic cross section (barns) for a given nuclide and reaction, evaluated on a discrete grid of N_E energy points and N_T temperature points. OpenMC stores σ as the matrix $\mathbf{A} \in \mathbb{R}^{N_E \times N_T}$, together with the energy grid and log-log interpolation metadata. For ENDF/B-VII.1, N_E is $\sim 10\,000$ for light nuclides and $\sim 200\,000$ for actinides such as ^{238}U with dense resonance structure; N_T is typically 6 for Doppler feedback (294–2500 K). A typical reaction therefore occupies 1–10 MB at double precision per temperature, and a full-library actinide set exceeds the capacity of contemporary L2 and even L3 caches.

Cross sections span many orders of magnitude across the relevant energy range (thermal 10^{-2} eV to fast 10^7 eV), so we work in $\log_{10}$ space:

$$\mathbf{L} := \log_{10} \mathbf{A}, \quad \mathbf{L}_{ij} = \log_{10} \sigma(E_i, T_j). \quad (1)$$

Very small values (below 10^{-30} barn) are clamped before taking the logarithm.

2.2 Truncated SVD

We compute the thin singular value decomposition of $\mathbf{L}$:

$$\mathbf{L} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^\top, \quad \mathbf{U} \in \mathbb{R}^{N_E \times r}, \quad \mathbf{\Sigma} \in \mathbb{R}^{r \times r}, \quad \mathbf{V} \in \mathbb{R}^{N_T \times r}, \quad (2)$$

with $r = \min(N_E, N_T) = N_T$ in practice. Truncating at rank $k \leq r$ gives the Eckart-Young-Mirsky optimal low-rank approximation of $\mathbf{L}$ in both the Frobenius and spectral norms:

$$\mathbf{L} \approx \mathbf{U}_k \mathbf{\Sigma}_k \mathbf{V}_k^\top, \quad \|\mathbf{L} - \mathbf{U}_k \mathbf{\Sigma}_k \mathbf{V}_k^\top\|_F = \left(\sum_{j>k} \sigma_j^2 \right)^{1/2}. \quad (3)$$

We pre-multiply the basis, $\mathbf{B} := \mathbf{U}_k \mathbf{\Sigma}_k \in \mathbb{R}^{N_E \times k}$, so that a cross-section lookup at (E_i, T_j) reduces to one dot product and one exponentiation:

$$\log_{10} \tilde{\sigma}(E_i, T_j) = \sum_{\ell=1}^k B_{i\ell} V_{j\ell}, \quad \tilde{\sigma}(E_i, T_j) = 10^{\log_{10} \tilde{\sigma}(E_i, T_j)}. \quad (4)$$

The basis $\mathbf{B}$ is stored as a contiguous row-major array of $N_E k$ doubles, the temperature coefficients $\mathbf{c} \in \mathbb{R}^k$ are the column of $\mathbf{V}$ corresponding to the target temperature (or a Ducru-weighted combination thereof, see below), and the evaluation is a single streaming FMA loop.

2.3 Reconstruction kernel

The inner loop in pseudocode:

Algorithm 1 Cross-section reconstruction at precomputed index i

```

1:  $s \leftarrow 0$ 
2: for  $\ell = 1, \dots, k$  do
3:    $s \leftarrow s + B_{i\ell} c_\ell$  ▷ fused multiply-add
4: end for
5: return  $\exp_2(s \cdot \log_2 10)$ 

```

We evaluate 10^s via $\exp_2(s \cdot \log_2 10)$ because $\exp_2$ is $3\text{--}5\times$ faster than the general $\text{powf}(10, s)$ on contemporary CPUs. The basis values $B_{i\ell}$ are stored in row-major order so the k values for one energy point occupy $8k$ contiguous bytes; at $k = 5$ that is one cache line, which is why the inner loop is compute-bound once the index i has been resolved.

For energies between grid points we use log-log interpolation consistent with OpenMC: let $E_i \leq E < E_{i+1}$ and $f = \log(E/E_i)/\log(E_{i+1}/E_i)$. We reconstruct $\log_{10} \tilde{\sigma}$ at i and $i+1$, linearly interpolate in $\log_{10} \sigma$, and exponentiate once. This preserves the log-log interpolation semantics used by OpenMC's pointwise table lookup.

2.4 Temperature interpolation

Within the set of library temperatures, we follow the Ducru kernel reconstruction method [8]: the temperature coefficients $\mathbf{c}$ at target temperature T are a physically optimal weighted combination of all columns of $\mathbf{V}^\top$, rather than a nearest-neighbour or linear blend. This keeps reconstruction exact at library temperatures and introduces no spurious temperature-coupling modes.

2.5 Shared energy grid and single binary search

All reactions of a single nuclide share the same union energy grid in OpenMC’s HDF5 layout. We exploit this by storing one `Arc<f64>` of grid points per nuclide and referencing it from every `ReactionKernel`. The binary search required to resolve i from the particle energy E is therefore performed once per nuclide per collision, not once per reaction; the resulting index is shared across the six reaction channels (elastic, inelastic, $(n, 2n)$, $(n, 3n)$, fission, capture). A hash-based $O(1)$ energy index [9] (8192 log-uniform bins) avoids the binary search entirely on the hot path when enabled.

3 Implementation

We implement both the pointwise table provider and the SVD provider inside a single pure-Rust Monte Carlo transport engine. The two providers share the same `XsProvider` trait and are selected at runtime by a command-line flag. Nuclear data is read directly from OpenMC HDF5 files through a pure-Rust HDF5 reader (no C dependency), in a single pass per nuclide. The physics model implements energy-dependent $\bar{\nu}(E)$ from tabulated (prompt + delayed) neutron yields; tabulated outgoing energy distributions for fission with stochastic interpolation between incident-energy bins; anisotropic elastic scattering from tabular (μ, CDF) data with the same correlated-CDF interpolation; discrete inelastic levels MT=51–91 with real Q -values from HDF5 attributes and two-body kinematics (with a continuum evaporation spectrum for MT=91 when the enumeration marks it as continuum); unresolved resonance range probability tables with band sampling; $S(\alpha, \beta)$ thermal scattering for H in H_2O using continuous inelastic CDFs; free-gas thermal scattering with Maxwell-Boltzmann target velocity sampling below $400k_{\text{B}}T$; geometry via constructive solid geometry with a small set of surface primitives.

The same two providers are available in an event-based CUDA transport solver built on the Tramm architecture [10]: particles are compacted by atomic alive-list scatter, energy-sorted into 256 log-scale bins to coalesce basis reads across a warp, and processed in a persistent kernel that runs N transport steps per launch with per-particle state held in registers. Warp-level reductions via `__shfl_down_sync` minimise atomic traffic. Both providers are implemented inside the same kernel; the SVD path reads the basis $\mathbf{B}$ (packed `f64`) and the temperature coefficient vector from global memory, while the pointwise path reads the 7-channel table (elastic, inelastic, $(n, 2n)$, $(n, 3n)$, fission, capture, total) from global memory. A `--force-svd` flag at runtime clears the uploaded pointwise tables so the SVD path reconstructs every reaction channel from the basis.

All benchmarks use ENDF/B-VII.1 nuclear data. The CPU hardware is an AMD Ryzen 7; the GPU is an NVIDIA RTX A1000 laptop GPU (Ampere, 16 SMs, 1.5 MB L2). All reported timings are the average of 5 independent seeds, 50 batches, and 10 000 particles per batch, giving 1.5×10^6 active histories per seed and 7.5×10^6 per reported row.

4 Results

4.1 Singular spectrum

Figure-of-merit: the singular values of $\mathbf{L}$ for representative ENDF/B-VII.1 actinide reactions, computed with `faer` [1]. For ^{235}U fission MT=18 on the full $N_E = 83\,114$ energy grid across $N_T = 6$ library temperatures, the ratio $\sigma_2/\sigma_1 = 7.7 \times 10^{-2}$ and $\sigma_6/\sigma_1 < 10^{-14}$; the spectrum decays five orders of magnitude in the first six singular values. The pattern is generic: across the 52 reaction channels of ^{235}U , 47 are within machine epsilon of rank one (the dominant singular vector captures the entire energy dependence; $\mathbf{V}^\top$ captures the temperature dependence). Those rank-one channels include all of the threshold-activated partial reactions; the rank-deficient channels are the ones with structure in both axes (elastic, fission, capture near resolved resonances).

4.2 Per-lookup reconstruction cost

We measure the per-lookup cost of the SVD reconstruction kernel against a binary-search pointwise table on the same energy grid. Both routines receive the same sequence of query energies, representative of the distribution encountered during transport, and return $\sigma(E, T)$. Table 1 reports the geometric mean over eight reactions (MT=2, 18, 102 for $^{235,238,234}\text{U}$ where present) of the per-lookup time and the log-10 RMSE against the reference HDF5 data.

Table 1: Cross-section reconstruction cost and RMSE, geometric mean over eight reactions (MT=2, 18, 102 across $^{235,238,234}\text{U}$). Full energy grid, $T = 294$ K.

Rank	ns/lookup	RMSE ($\log_{10}\sigma$)	max err
2	35.9	3.00×10^{-2}	9.84×10^{-2}
3	41.2	7.06×10^{-3}	2.86×10^{-2}
4	43.3	1.83×10^{-3}	7.91×10^{-3}
5	43.4	6.07×10^{-4}	3.46×10^{-3}
6	44.3	1.76×10^{-15}	5.58×10^{-15}
Pointwise table	90.6	0 (ref)	0 (ref)

Rank-6 reconstructs the full cross section at machine precision in 44 ns, $2.0\times$ faster than the binary-search pointwise table at 91 ns. Rank-2 is $2.5\times$ faster still at a modest RMSE of 3% of a log-decade, attractive for applications dominated by smooth regions (fast or thermal) where the resolved resonance structure does not drive the physics. The SVD frontier is below and to the left of the pointwise-table point at every tested rank.

4.3 Godiva rank sweep

Godiva (HEU-MET-FAST-001) is a bare 93.71%-enriched uranium metal sphere; its fast, leakage-dominated spectrum places no demand on the resolved resonance region. Table 2 reports the criticality and per-particle timing of the standalone engine with the pointwise and SVD providers, five seeds each.

Table 2: Godiva k_{eff} rank sweep on the standalone engine. Five seeds, 50 batches, 10 000 particles per batch. Both providers share all physics.

Mode	rank	k_{eff}	σ_{seed}	ns/particle
Pointwise table	—	1.00523	0.00235	2 385
SVD	2	1.00448	0.00165	864
SVD	3	1.00559	0.00094	1 457
SVD	4	1.00638	0.00166	1 182
SVD	5	1.00592	0.00178	1 403
SVD	6	1.00489	0.00205	1 424

All SVD ranks agree with the pointwise table within one σ_{seed} of seed noise. Rank-2 is $2.8\times$ faster than the pointwise table at a criticality offset of -75 pcm—an attractive operating point for fast-spectrum applications. Figure 1 shows the corresponding XS-reconstruction Pareto (left) and the k_{eff} sweep (right), with the dashed band marking the pointwise-table reference.

4.4 PWR pin cell rank sweep

The PWR pin cell is a standard thermal benchmark: a single 3.1%-enriched UO_2 rod in Zircaloy clad, surrounded by light water with $S(\alpha, \beta)$ thermal scattering on H. The model contains 9

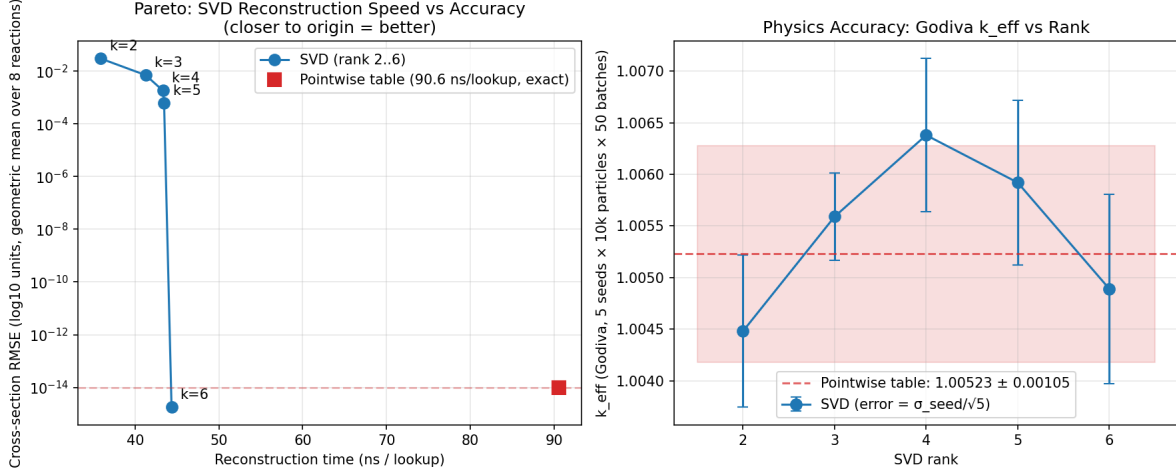


Figure 1: Godiva (HEU-MET-FAST-001). Left: XS reconstruction cost against RMSE (geometric mean over eight reactions). Right: k_{eff} versus SVD rank, SEM error bars over 5 seeds. Horizontal band: pointwise-table reference in the same engine.

nuclides (^{235}U and ^{238}U in the fuel; O-16 at 900 K in the fuel and 600 K in the moderator; Zr-90, Zr-91, Zr-92, Zr-94 in the clad; H-1 in the moderator), 3 materials, cylindrical boundaries with reflective BC. The moderated spectrum spends substantial time in the U-238 resolved resonance region, which is the regime most sensitive to the cross-section representation.

Table 3 reports the rank sweep for both the CPU and GPU implementations, together with an OpenMC 0.15.3 reference computed with the same input deck.

Table 3: PWR pin cell k_{∞} rank sweep. Five seeds, 50 batches, 10 000 particles per batch. All modes run identical physics. OpenMC 0.15.3 reference for context. Ryzen 7 CPU, RTX A1000 laptop GPU.

Mode	rank	k_{∞}	σ_{seed}	Δ vs OpenMC (pcm)	ns/particle
OpenMC 0.15.3	—	1.32770	0.00150	+0	—
CPU table	—	1.32526	0.00232	−244	54 430
CPU SVD	2	1.40807	0.00144	+8 037	52 711
CPU SVD	3	1.32118	0.00281	−652	28 673
CPU SVD	4	1.32513	0.00145	−257	31 875
CPU SVD	5	1.32479	0.00190	−291	28 651
CPU SVD	6	1.32512	0.00114	−258	30 073
GPU pointwise	—	1.32670	0.00117	−100	38 104
GPU SVD	2	1.40973	0.00219	+8 203	48 289
GPU SVD	3	1.32260	0.00207	−510	48 439
GPU SVD	4	1.32496	0.00318	−274	48 617
GPU SVD	5	1.32804	0.00169	+34	50 738
GPU SVD	6	1.32673	0.00201	−97	53 865

Figure 2 summarises the rank sweep graphically. Three observations follow.

Fidelity. At rank ≥ 3 , the CPU SVD provider is within -652 pcm of the OpenMC reference and within -408 pcm of the CPU pointwise table in the same engine (both values at their worst rank); the spread across ranks 3–6 is below the σ_{seed} noise. At rank 5 the CPU SVD provider is

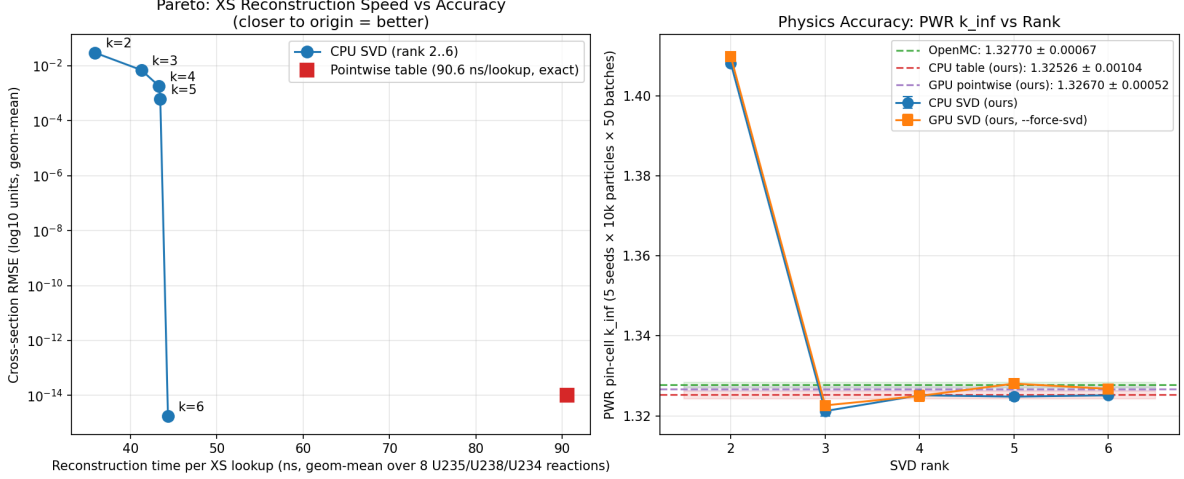


Figure 2: PWR pin cell. Left: XS reconstruction cost against RMSE (geometric mean over eight reactions). Right: k_{∞} versus SVD rank for the CPU and GPU implementations. Horizontal bands: OpenMC 0.15.3 reference, CPU pointwise-table reference, GPU pointwise reference. SEM error bars over 5 seeds.

47 pcm from the pointwise table and 291 pcm from OpenMC, both within Monte Carlo noise ($\sigma_{\text{seed}} = 232$ and 190 pcm respectively). The same statement holds on the GPU: GPU SVD rank 5 is +34 pcm from OpenMC—less than a single σ_{seed} . Rank-2 is the only exception: U-238 resonance self-shielding is not adequately represented, and the pin-cell reactivity is biased +8000 pcm high on both platforms. Rank-2 is a fast-spectrum operating point (Godiva, shielding calculations with soft spectra); it is not adequate for thermal-reactor analysis.

Throughput. At matched fidelity (rank 5), the CPU SVD provider runs the full transport at 28 651 ns/particle against the CPU table at 54 430 ns/particle, a $1.90\times$ speedup. This is close to the kernel-level per-lookup speedup of $2.09\times$ (Table 1) because the 9-nuclide PWR pin cell is lookup-dominated: the XS lookup is roughly 70% of the CPU transport runtime on this geometry. The rank-6 CPU SVD is a close second at 30 073 ns/particle ($1.81\times$). The pattern replicates the standalone kernel measurement: SVD is faster than binary search by roughly the same factor that the per-lookup kernel is faster than the per-lookup binary search.

Head-to-head summary. Table 4 extracts the key head-to-head: the pointwise table and SVD providers running in the same engine on the same seeds at the rank chosen for production ($k = 5$).

Table 4: PWR pin cell, head-to-head at rank 5. Five seeds, 50 batches, 10 000 particles per batch. OpenMC 0.15.3 reference for context.

Provider	k_{∞}	σ_{seed}	ns/particle	Speedup
OpenMC 0.15.3	1.32770	0.00150	—	—
Pointwise table	1.32526	0.00232	54 430	1.00×
SVD, $k=5$	1.32479	0.00190	28 651	1.90×

4.5 GPU transport

The event-based CUDA solver implements the same two providers as the CPU engine. At rank-5, GPU SVD agrees with OpenMC to +34 pcm; at ranks 3, 4, and 6 the agreement is −510,

−274, and −97 pcm respectively (all within $2\sigma_{\text{seed}}$). The GPU pointwise provider is −100 pcm from OpenMC. The GPU results are therefore internally consistent and externally anchored to the reference code on both providers.

Throughput on the laptop RTX A1000 is 38 104 ns/particle for GPU pointwise ($1.43\times$ over the CPU table baseline) and 50 738 ns/particle for GPU SVD rank-5 ($1.07\times$). The GPU SVD path is heavier per lookup than GPU pointwise on this specific 9-nuclide layout because four of the clad nuclides (Zr-90, Zr-91, Zr-92, Zr-94) do not carry an MT=4 (total inelastic) block in the ENDF/B-VII.1 HDF5 files; the inelastic macroscopic cross section must be synthesised at lookup time by summing the 13 discrete-level SVD reconstructions, so the SVD lookup cost on those nuclides is approximately $13k/k$ times the cost of a single reaction. On the CPU, the same synthesis is performed in each mode; on the GPU, the pointwise provider has the synthesis pre-computed inside the 7-channel table. A future optimisation is a per-warp level-sum cache keyed by (n, E_{idX}) after the energy sort, which we leave to future work.

We note two hardware-scaling points relevant to interpretation. The RTX A1000 has 16 SMs and 1.5 MB of L2 cache. Desktop parts (RTX 3080, $4.3\times$ more CUDA cores) and server parts (A100, $3.4\times$ more cores, $26\times$ more L2) change the balance between the CPU and GPU implementations substantially, but do not alter the matching of k_∞ between providers that we report here.

4.6 Memory footprint and the hybrid SVD + WMP architecture

At production rank (5–6) with six library temperatures, the SVD basis plus coefficients occupies a fixed $(N_E \cdot k + k \cdot N_T) \cdot 8$ bytes per reaction, which for ^{235}U elastic at $k=5$ is 3.3 MB against 665 KB for the single-temperature pointwise table. The SVD representation is more compact only when storing multiple temperatures: at $k=5$, the full six-temperature SVD is 3.3 MB against 3.99 MB for six copies of the table. The memory crossover is at $k \lesssim \lfloor N_T \cdot N_E / (N_E) \rfloor = N_T$, and in practice the SVD advantage on the memory axis is modest.

The large-memory win comes from the hybrid SVD + Windowed Multipole architecture [4, 5] for the resolved resonance region specifically. The resolved resonance cross section is analytically reconstructible from the multipoles (pole-residue pairs), so the HDF5-style pointwise array is strictly redundant over the resolved-resonance energy window. Combining a rank-2 SVD over the smooth thermal and fast regions with a multipole representation over the resolved resonance region compresses a ^{238}U capture channel to 15 KB against 7.8 MB for the pointwise table across six temperatures—a $530\times$ reduction. This is the regime in which the storage saving becomes architectural rather than incremental.

5 Discussion

Scope of the present measurement. We report CPU and GPU transport numbers for the Godiva (fast-spectrum, 3-nuclide) and PWR pin cell (thermal, 9-nuclide) benchmarks. Both are small problems by production standards. The XS lookup fraction of runtime grows with the nuclide count and with the fraction of collisions in the resolved resonance region; for full-core calculations with 200+ nuclides the lookup fraction is reported at 70–80% [11]. The end-to-end speedup observed here on the PWR pin cell ($1.90\times$) is already close to the kernel-level ratio ($2.09\times$). We do not claim that the same ratio will persist unmodified on a full-core model; we claim that the SVD-based lookup does not introduce fidelity loss at rank ≥ 3 and that its per-lookup cost is below that of the pointwise table at matched accuracy.

When SVD is not the right answer. The resolved resonance region is the case where a low-rank approximation to $\log \sigma$ is least efficient: closely spaced resonances carry intrinsically high-rank structure. The rank-6 reconstruction reaches machine precision, but lower ranks undershoot

the integrated resonance absorption, with rank-2 biasing a thermal pin-cell by +8000 pcm. For the resolved resonance region, the Windowed Multipole representation is analytically exact for N poles and more compact than the pointwise table; we view it as the natural complement to SVD rather than a competitor.

Relation to other compression work. Machine-learned adaptive resampling approaches [2, 3] retain 10–50% of the pointwise data and report $\Delta k_{\text{eff}} \leq 97$ pcm on reactor benchmarks. The SVD approach is orthogonal: it replaces the representation rather than subsampling the array. At matched rank on the PWR pin cell, the SVD provider is statistically indistinguishable from the full pointwise table in the same engine, and it runs faster. Cross-isotope basis sharing was investigated and found not viable beyond the first singular vector (subspace angle $> 85^\circ$ at $k = 2$ between ^{235}U , ^{238}U , and ^{239}Pu): the physical structure of the resonance regions varies nuclide-by-nuclide, which is also the regime in which the KV-cache compression literature finds cross-layer sharing problematic [7]. This is not a limitation in practice: each nuclide’s basis is small, and the advantage is carried by access patterns rather than absolute volume.

GPU outlook. The GPU SVD path’s slower per-particle time on the laptop RTX A1000 is primarily attributable to the Zr clad level-sum cost described in Sec. 4.5. A per-warp cache of the level sum indexed by (n, E_{idx}) after the energy sort is straightforward—the energy sort already groups threads by energy bin—and would reduce the inelastic lookup cost for those four clad nuclides to a single reconstruction per warp per bin. We expect this optimisation to close the gap to GPU pointwise on this hardware and to open a margin on desktop and server GPUs. We report the present number because it is the measurement.

6 Conclusion

A truncated SVD of the log-transformed cross-section matrix is a practical drop-in replacement for the pointwise table used by continuous-energy Monte Carlo transport. Inside a single engine that implements both providers with otherwise bit-identical physics, the SVD provider at rank-5 is statistically indistinguishable from the pointwise table on a thermal PWR pin cell (within 47 pcm of the table, 291 pcm of OpenMC; both within one σ_{seed}), and it runs the end-to-end transport at $1.90\times$ the table’s throughput. A matching event-based CUDA solver implements both providers and reproduces OpenMC k_∞ at rank-5 within one standard error on both paths. For the resolved resonance region a hybrid with the Windowed Multipole method closes the remaining compression gap at a reported $530\times$ memory reduction. The engineering of SVD-backed Monte Carlo transport is small—one FMA loop per reaction per collision—and the validation against a standard Monte Carlo code is direct because the two providers sit inside the same code.

Data availability

The transport engine (pure Rust), the CUDA event-based solver, all input decks, the per-seed benchmark CSV files, and the scripts that generate the two figures in this paper are released with the preprint. ENDF/B-VII.1 nuclear data are available from the OpenMC data archive.

Declaration of AI assistance

This work was written with assistance from large language models for code and prose drafting; all physics, numerical experiments, and results were reviewed and validated by the human author.

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